

# Pitambar Acharya

Ph.D. Candidate in Applied Mathematics

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*Applied mathematician and Ph.D. candidate specializing in nonlinear partial differential equations and mathematical modeling, with research interests spanning PDE analysis, computational mathematics, data science, and machine learning. Experienced educator with a broad teaching portfolio across Precalculus, Statistics, Calculus, Linear Algebra, and other foundational undergraduate mathematics courses.*

## RESEARCH INTERESTS

Partial Differential Equations • Mathematical Biology & Chemotaxis Models • Data Science • Machine Learning • Scientific Computing

## EDUCATION

|                                    |                                                                                                                                                                                                                                                                                                                 |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aug 2022 – Expected<br>Summer 2027 | <b>PhD, Applied and Computational Mathematics</b><br><i>University of Alabama at Birmingham, Birmingham, AL</i><br>Dissertation: <i>Global Existence of Solutions on the Quarter Plane for a Chemotaxis Model with Logistic Growth and Logarithmic Sensitivity</i><br>Advisor: <a href="#">Prof. Yanni Zeng</a> |
| Aug 2025                           | <b>MS, Mathematics</b><br><i>University of Alabama at Birmingham, Birmingham, AL</i>                                                                                                                                                                                                                            |
| Jan 2020 – Jul 2022                | <b>MS, Applied Mathematics and Statistics</b> (Biostatistics Track)<br><i>Florida Atlantic University, Boca Raton, FL</i>                                                                                                                                                                                       |
| 2012 – 2014                        | <b>MSc, Mathematics</b><br><i>Tribhuvan University, Kirtipur, Kathmandu, Nepal</i>                                                                                                                                                                                                                              |
| 2010 – 2012                        | <b>MSc, Physics</b><br><i>Tribhuvan University, Kirtipur, Kathmandu, Nepal</i>                                                                                                                                                                                                                                  |
| 2007 – 2010                        | <b>BSc, Physics</b> (Minor: Mathematics and Statistics)<br><i>Trichandra College, Ghantaghar, Kathmandu, Nepal</i>                                                                                                                                                                                              |

## TEACHING EXPERIENCE

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aug 2022 – Present | <b>Instructor of Record</b><br><i>University of Alabama at Birmingham, Birmingham, AL</i> <ul style="list-style-type: none"><li>• Precalculus: Algebra/Trigonometry (MA 107 QL) – Summer 2026</li><li>• Calculus I (MA 125/125L) – Spring 2026</li><li>• Precalculus: Trigonometry (MA 106) – Fall 2026, Spring 2025, Fall 2024</li><li>• Introduction to Statistics (MA 180) – Fall 2025, Summer 2025</li><li>• Precalculus: Algebra (MA 105) – Spring 2024, Fall 2023, Spring 2023</li><li>• Intermediate Algebra (MA 102) – Fall 2022</li></ul> |
| Jan – May 2022     | <b>Instructor of Record</b><br><i>Florida Atlantic University, Boca Raton, FL</i> <ul style="list-style-type: none"><li>• Methods of Calculus – Spring 2022</li></ul>                                                                                                                                                                                                                                                                                                                                                                              |

Jan 2020 – Jul 2022

### Teaching Assistant

Florida Atlantic University, Boca Raton, FL

- Calculus II (MAC 2312) – Spring 2021, Fall 2020
- Calculus I (MAC 2311) – Spring 2020
- Engineering Mathematics I (MAP 3305) – Summer 2020
- Intermediate & College Algebra: quizzes and midterms – Fall 2021
- Tutor, Engineering Mathematics and Mathematics for Liberal Arts – Summer 2021

Jun 2015 – Dec 2019

### Lecturer

Himalaya College of Engineering (Tribhuvan University), Nepal

- Probability and Statistics for BE and BSc CSIT programs
- Engineering Physics (lectures and labs): Civil, Architecture, Computer, and Electronics Engineering students

## PUBLICATIONS

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### Peer-Reviewed Journal Articles

- [1] K. Acharya and P. Acharya. “Adaptive Filtering via Canonical Systems with Time-Varying Hamiltonians.” *International Journal of Advances in Engineering Sciences and Applied Mathematics* (Springer Nature), 18, 20–30 (2026). DOI: 10.1007/s12572-026-00421-5
- [2] E. P. Poudel, P. Acharya, J. Kafle, and S. Khadka. “On One-Dimensional Advection–Diffusion Equation with Variable Diffusivity.” *Applied Mathematics: A Journal of Chinese Universities* (Springer Nature), 41, 181–193 (2026). DOI: 10.1007/s11766-026-5149-9
- [3] D. Dulal, M. Gallagher, C. Navasca, P. Acharya and N. Skowronski. “Leveraging Advanced Machine Learning to Predict Turbulence Dynamics From Temperature Observations at an Experimental Prescribed Fire.” *IEEE Access*, 14, 112481–112498 (2026) DOI:10.1109/ACCESS.2026.3715743.
- [4] E. P. Poudel, P. Acharya, J. Kafle, and S. Khadka. “Modeling and Analysis of Two-Dimensional Contaminant Dispersion in Turbulent Atmospheric Flows.” Submitted to *Electronic Journal of Differential Equations*.
- [5] D. Bastola, P. Acharya, and D. Dulal. “Predicting Anemia Among Under-Five Children in Nepal Using Machine Learning and Deep Learning.” Submitted to *Public Health Nutrition*. Preprint
- [6] K. Acharya, P. Acharya, and D. Bastola. “Random Schrödinger Operators on Data Graphs: A Data Science Perspective on Robust Spectral Embedding.” Manuscript in preparation.

## ADDITIONAL RESEARCH AND PROJECTS

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- [1] P. Acharya, L. Zeng, and M. Islam. “Comparative Analysis of Data Mining Techniques for Customer Segmentation and Anomaly Detection.” [Graduate final project](#).
- [2] M. Islam, H. Akther Samia, P. Acharya. “Bin Packing Problem with Machine Learning Techniques.” [View Project](#)
- [3] P. Acharya, O. Francis, and M. Islam. “Deep Q-Learning for Optimizing Bin Packing Problem: A Deep Reinforcement Learning Approach.”

## PRESENTATIONS

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|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aug 2025 | “Learning the Solution Operator of the Heat Equation.”<br><i>Summer School on Scientific Machine Learning, Brin Mathematics Research Center, University of Maryland, College Park</i> |
| Feb 2025 | “On One-Dimensional Advection–Diffusion Equation with Variable Diffusivity.”<br><i>MAA–FTYCMA Joint Conference, Embry-Riddle Aeronautical University, Daytona Beach</i>               |
| May 2022 | “Bayesian Model Averaging: An Overview.”<br><i>MS Presentation, Florida Atlantic University, Boca Raton, FL</i>                                                                       |

## AWARDS & HONORS

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|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Apr 2025</i>       | James Robert Ward, Jr. Memorial Endowed Award (\$1,500) for excellence in Graduate Studies in Differential Equations / Mathematical Biology – UAB |
| <i>Mar 2025</i>       | Travel Award (\$563.61) – Ohio River Analysis Meeting, University of Cincinnati                                                                   |
| <i>Jun–Jul 2024</i>   | Professional Development Stipend (\$1,000) – Blazer Core Curriculum, UAB                                                                          |
| <i>Jul 2023</i>       | Travel Award (\$1,066) – Texas Summer School in Mathematical Physics, Texas State University                                                      |
| <i>2022 – Present</i> | Graduate Scholarship / Teaching Assistantship – UAB                                                                                               |
| <i>2020 – 2022</i>    | Graduate Scholarship / Teaching Assistantship – FAU                                                                                               |
| <i>2012 – 2014</i>    | MS Fellowship, Central Department of Mathematics, Tribhuvan University                                                                            |
| <i>2010 – 2012</i>    | MS Fellowship, Central Department of Physics, Tribhuvan University                                                                                |

## CONFERENCES & WORKSHOPS ATTENDED

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### Selected Conferences (recent)

- Fourth Annual Joint Alabama–Florida Conference on Differential Equations, Dynamical Systems & Applications (JAF DEDS) – FIU, May 2026
- 37th Annual UA System Applied Mathematics Meeting – UAH, Nov 2025
- 3rd UCG Virtual PDE Conference – UNC Greensboro, Oct 2025
- Ohio River Analysis Meeting (ORAM 14) – University of Cincinnati, Mar 2025
- Southeastern Analysis Meeting (SEAM 41) – University of South Florida, Mar 2025
- Modeling and Simulations in Fluids – Brown University (ICERM), Sep 2024
- Joint Mathematics Meetings (JMM) – Boston, MA, Jan 2023

### Summer Schools & Workshops

- Summer School: Scientific Machine Learning – Brin MRC, University of Maryland, Aug 2025
- Texas Summer School in Mathematical Physics – Texas A&M University, Jun 2025
- Texas Summer School in Mathematical Physics – Texas State University, Jul 2023
- CIMPA Summer School: Data Visualization, Modeling and Mathematical Tools – Tribhuvan University, Nepal, May 2023 (*Organizing Committee Member*)

## LEADERSHIP & SERVICE

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| <i>Aug 2024 – Present</i>  | <b>Secretary</b> – SIAM Student Chapter, UAB                                                                                                                                                                                                                                                                      |
| <i>Aug 2022 – Present</i>  | <b>Executive Member</b> – Nepalese Students Association, UAB                                                                                                                                                                                                                                                      |
| <i>Jul 2019</i>            | <b>Deputy Leader</b> – International Mathematics Olympiad (IMO), Bath, United Kingdom                                                                                                                                                                                                                             |
| <i>Jul 2017</i>            | <b>Deputy Leader</b> – International Mathematics Olympiad (IMO), Rio de Janeiro, Brazil                                                                                                                                                                                                                           |
| <i>Feb 2017 – Dec 2019</i> | <b>Vice President</b> – Mathematical Association of Nepal (MAN) <ul style="list-style-type: none"><li>• Trained high school students for the National Mathematics Competition</li><li>• Conducted examinations and prepared students for IMO</li><li>• Organized national and international conferences</li></ul> |
| <i>Aug 2020 – 2021</i>     | <b>Vice President</b> – Nepalese Students Association, FAU                                                                                                                                                                                                                                                        |

## VOLUNTEER WORK

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- Alabama Statewide High School Mathematics Contest – Feb 2026
- Math Talent Search Grader – Oct 2025
- Alabama Statewide Math Ciphering Contest – Apr 2025
- High School Science Exhibition Judge, Birmingham, AL – Jul 2023
- Kangaroo Mathematics Competition, UAB – Mar 2023
- Math Day Proctoring and Grading, FAU – Jan 2021
- High School Science Olympiad, FAU – Feb 2020
- Blood Donation Program, Kanti Children's Hospital, Kathmandu, Nepal
- Donated stationery and sports materials for cancer-affected children, Kanti Hospital, through Astitwo Foundation, Nepal

## TECHNICAL SKILLS

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|------------------------------|-------------------------------------------------------------------|
| <b>Programming Languages</b> | Python, R, MATLAB, SQL, Minitab                                   |
| <b>Python Libraries</b>      | Pandas, NumPy, Matplotlib, SciPy, Scikit-learn, TensorFlow, Keras |
| <b>Deep Learning</b>         | PyTorch (building and training neural networks)                   |
| <b>Tools &amp; Platforms</b> | Git, Linux, Jupyter Notebook, AWS, GCP                            |
| <b>Statistical Software</b>  | R (ggplot2), Minitab                                              |

## PROFESSIONAL MEMBERSHIPS

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- American Mathematical Society (AMS)
- Society for Industrial and Applied Mathematics (SIAM)
- Nepal Mathematical Society (NMS)
- Nepal Physical Society (NPS)
- Association of Nepalese Mathematicians in America (ANMA)

## PROFESSIONAL DEVELOPMENT

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- CTL LIFT Teaching Certificate – [Center for Teaching and Learning, UAB](#)
- Developing Great Presentations – UAB Learning and Development, May 2025
- Flex/Online Teaching Orientation – Centre for Online and Continuing Education (COCE), FAU
- Teaching Assistant Basics for Online Courses – COCE, FAU
- Inquiry-Based Learning Workshop – Department of Mathematical Sciences, FAU
- CRLA Certification Levels 1 & 2 – Mathematics Learning Center, FAU
- 14th Annual iLead Conference – Student Union Hall, FAU

## LANGUAGES

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- English (*professional proficiency*)
- Nepali (*native*)
- Hindi (*proficient*)

## REFERENCES

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**Dr. Yanni Zeng**

Professor

University of Alabama at Birmingham

[ynzeng@uab.edu](mailto:ynzeng@uab.edu) (205) 934-2154**Dr. Hongwei Long**

Professor

Florida Atlantic University

[hlong@fau.edu](mailto:hlong@fau.edu) (561) 297-0810**Dr. Yulia Karpeshina**

Professor; Director, Graduate Program

University of Alabama at Birmingham

[karpeshi@uab.edu](mailto:karpeshi@uab.edu) (205) 934-2154**Dr. Elena Kravchuk**

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